

Marketing Campaign Analysis

Technical Report — Full Project Documentation

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Datasets	marketing_campaign.csv — 2,240 European company customers (2,212 after cleaning) Datafiniti Amazon Reviews — 28,332 reviews (25,673 after cleaning)
Stack	Python (pandas, scikit-learn, mlxtend, NLTK) · PostgreSQL · SQLAlchemy · Tableau Public
Phases	1 EDA · 2 Cleaning & DB Loading · 3 Campaign Analysis · 4 CLV Prediction · 5 Sentiment Analysis · 6 Market Basket · Dashboard
Business questions	1. Which campaigns generate positive ROI and which customers respond? 2. Which customers are most valuable and likely to respond to future campaigns? 3. Which product categories are purchased together and how does this vary by CLV tier? 4. What do customers say about products and which sentiment model best identifies complaints?

Important framing — two structurally disjoint datasets. The campaign and Amazon reviews datasets share no customer IDs, product IDs, or any other join key. The Amazon reviews were brought in specifically to demonstrate NLP skills since the campaign dataset contains no text data. Phase 5 sentiment findings are standalone insights — they cannot be tied back to specific campaign customers, CLV tiers, or response rates. Any conclusion linking the two datasets would be a logical error.

Phase 1 — Exploratory Data Analysis

Objective

Understand the structure, quality, and key distributions of both datasets before any cleaning or modelling. Every data quality issue found here drives a specific fix in Phase 2.

What we did

Campaign dataset (2,240 customers, 29 columns). Profiled demographics (age, education, income, marital status, household composition), spending across 6 product categories, purchase channel preferences, and campaign acceptance rates. Computed summary statistics and plotted distributions for all key variables.

Reviews dataset (28,332 reviews, 65 unique products). Examined the star-rating distribution, review text length, and duplicate patterns. Identified a severe class imbalance (90.2% positive, 4.3% neutral, 5.6% negative) that would need to be addressed in Phase 5.

Duplicate check on both datasets. Confirmed zero duplicate IDs in the campaign dataset. Found 1,669 reviews sharing the same (product, review text) pair — these would inflate the positive sentiment class during model training if not removed.

Key findings

Finding	Detail
Wines dominate spending	Wines account for \$680,816 total — almost 2× the next category (Meat at \$373,968)
Store is the top channel	12,970 store purchases vs 9,150 web, 5,963 catalogue, 5,208 deals
Latest campaign outperforms	14.91% acceptance vs 1.34–7.46% for the 5 historical campaigns
Review class imbalance	90.2% positive, 4.3% neutral, 5.6% negative — standard classifier would predict positive for everything
Age reference year	Originally hardcoded to 2015; corrected to derive empirically from $\max(\text{Dt_Customer}) = 2014$
Spending labels bug	Spending category labels were assigned after <code>sort()</code> , scrambling the chart. Fixed by renaming before sorting.

Data quality issues flagged for Phase 2

Issue	Detail	Action
Constant columns	Z_CostContact and Z_Revenue: std = 0	Drop both
Age outliers	3 customers with Year_Birth ≤ 1900	Drop
Income outlier	1 customer with Income = \$666,666	Drop
Missing income	24 rows (1.07%) — 4.2% response rate vs 15% for the rest	Drop, flag bias
Marital status junk	YOLO, Absurd, Alone categories	Consolidate into Single
Education mislabel	'2n Cycle' mapped to Secondary — incorrect for EU system	Remap to Undergraduate
Review duplicates	1,669 (product, text) pairs	De-duplicate before Phase 5
Very short reviews	2,538 reviews under 20 characters	Filter from sentiment training

Target leakage flag for Phase 4. Response is the target for propensity modelling. Any 'campaigns accepted' feature must use only AcceptedCmp1–5 — never include Response in features predicting itself. Built

Total_Hist_Campaigns_Accepted as the leakage-safe version.

Phase 2 — Data Cleaning & Database Loading

Objective

Apply all data quality fixes identified in Phase 1, engineer downstream features, and load a normalised PostgreSQL database with enforced primary key and foreign key constraints that all subsequent phases query.

Why a relational database?

The original flat file mixes demographics, spending, channel preferences, and campaign responses into a single wide table. Splitting it into 4 relational tables (customers, spending, channels, campaigns) makes downstream SQL queries cleaner and more readable, mirrors how real marketing databases are structured, and enforces referential integrity at the database level rather than relying on application-level checks.

Cleaning steps

Step	Action	Rows / Cols affected
Constant columns	Dropped Z_CostContact and Z_Revenue (std = 0)	–2 columns
Missing income	Dropped 24 rows — behaviourally biased (4.2% vs 15% response rate)	–24 rows
Age outliers	Dropped Year_Birth ≤ 1900	–3 rows
Income outlier	Dropped Income = \$666,666	–1 row
Education	'2n Cycle' → 'Undergraduate' (second cycle of higher education, not high school)	Renamed
Marital status	YOLO / Absurd / Alone → Single	Consolidated
Date parsing	Dt_Customer parsed with dayfirst=True (DD-MM-YYYY format)	Type fix
Age derivation	Age = max(Dt_Customer).year – Year_Birth (empirical, not hardcoded)	New column
Review duplicates	Removed (product_name, review_text) duplicates — 1,261 excess copies	–1,261 rows
Short reviews	Filtered reviews under 20 characters	–1,398 rows

Feature engineering

Feature	Definition	Used in
TotalSpend	Sum of all 6 category spend columns	Phases 3, 4, 6
TotalPurchases	Sum of all 4 channel purchase columns	Phase 3
TotalChildren	Kidhome + Teenhome	Phase 4
HasChildren	Binary: TotalChildren > 0	Phase 4
TenureDays	Days from Dt_Customer to max(Dt_Customer)	Phase 4
TotalHistCampaignsAccepted	Sum of AcceptedCmp1–5 (excludes Response)	Phases 3, 4
AvgSpendPerPurchase	TotalSpend / TotalPurchases (0 if no purchases)	Phase 3
Sentiment label	Negative (≤ 2 stars) / Neutral (3 stars) / Positive (4–5 stars)	Phase 5

Database schema

The schema was executed with real PRIMARY KEY and FOREIGN KEY constraints before data insertion, so referential integrity is enforced at the database level. Load order respects FK direction: customers (parent) must be inserted before spending, channels, and campaigns (children).

Table	Rows	Primary key	Role
customers	2,212	customer_id	Parent — demographics + tenure
spending	2,212	customer_id	Spend per category (FK → customers)
channels	2,212	customer_id	Web / catalogue / store / deals (FK → customers)
campaigns	2,212	customer_id	5 historical + Response (FK → customers)
reviews	25,673	—	Amazon reviews (standalone — no FK to campaign tables)

Sample bias note. The 24 missing-income customers dropped had a 4.2% response rate vs 15% for the rest. The remaining dataset tilts slightly toward more engaged customers. This bias carries into Phase 4 — the propensity model is calibrated on a sample that is, on average, more responsive than the true population.

Phase 3 — Campaign Performance Analysis

Objective

Use SQL queries against the normalised database to answer the first business question: which campaigns are generating the most response, which customers respond, and how should this inform channel investment and future targeting?

Why SQL-first analysis?

The relational database structure makes aggregation, filtering, and joining across customer dimensions natural SQL operations. Computing acceptance rates, channel totals, and responder profiles in SQL — rather than in pandas after loading the full flat file — is more scalable, produces self-documenting queries, and lets the database engine optimise the aggregations. All 23 queries were saved to `sql/queries/` for audit.

Why break-even analysis instead of ROI?

The source dataset includes `Z_CostContact` (\$3) and `Z_Revenue` (\$11) — both are constants, the same value for every customer, and were dropped in Phase 2 as data-quality artefacts. With no real cost or revenue data, computing an 'actual ROI' would be fabrication. The defensible alternative is a break-even analysis: given each campaign's observed acceptance rate, what minimum revenue-per-response to cost-per-contact ratio would make it profitable? This answers the same business question honestly.

Formula: $\text{break-even ratio} = 1 / \text{acceptance_rate}$. A campaign with 15% acceptance breaks even when revenue per response $\geq 6.6\times$ the cost per contact.

Key findings

Finding	Detail
Latest campaign dominates	15.05% acceptance vs 1.36–7.41% for campaigns 1–5. Only profitable one under any realistic cost/revenue scenario.
Campaign 2 is dramatically inefficient	1.36% acceptance = $73.7\times$ break-even ratio. Would need revenue $73.7\times$ the contact cost to break even.
Past acceptance = strongest response signal	Responders accepted 0.99 historical campaigns on average vs 0.18 for non-responders — a $5.5\times$ gap.
Responders spend 82.6% more	Avg total spend \$986 for responders vs \$540 for non-responders.
Catalog is the luxury channel	High spenders use catalog $20\times$ more than low spenders (5.66 vs 0.29 avg purchases). Strongest channel signal in the data.
PhD customers respond at 21%	vs 3.7% for Basic education — largest demographic gap. Behavioural features still discriminate more sharply.
Web has high visits, low intent	5.3 visits/month average, but only ~ 4 web purchases over 2 years. Not a conversion gap — low per-visit purchase intent.

Phase 4 inputs from this phase

Past campaign acceptance, total spend, catalog purchases, income, and recency emerged as the strongest behavioural features separating responders from non-responders. `Total_Hist_Campaigns_Accepted` was built here and confirmed leakage-safe (excludes Response). The latest campaign's $2\times$ acceptance rate vs historical campaigns suggested its targeting and offer design are worth learning from in the propensity model.

Phase 4 — Customer Lifetime Value & Response Propensity

Objective

Train two separate model families: a regression to predict customer total spend (CLV proxy), and a classifier to predict whether a customer will accept the next campaign. Combine both into a targeting priority matrix.

CLV regression — leakage prevention

The regression target is `total_spend`. The following features were explicitly excluded to prevent target leakage:

- All individual category spend columns (`mnt_wines`, `mnt_fruits`, ...) — they sum directly to `total_spend`.
- All purchase count columns (`num_web_purchases`, `num_catalog_purchases`, etc.) — purchase counts and spend measure the same behaviour in different units. Including them would inflate R^2 without real predictive ability.
- `avg_spend_per_purchase` — derived from `total_spend`.

The honest test: can we predict customer value from who they are and how they engage, without already knowing how much they buy? Features used: `age`, `income`, `total_children`, `has_children`, `tenure_days`, `recency`, `num_web_visits_month`, `total_hist_campaigns_accepted`, plus one-hot encoded education and marital status.

Why one-hot encode education instead of ordinal? The dataset doesn't have a clean ordinal education structure — the spacing between Undergraduate, Graduation, Master, and PhD is not meaningfully equal. One-hot encoding lets the model learn the actual spend patterns per education level without imposing false ordering.

Regression results

Model	MAE	RMSE	R ²	Notes
Baseline (mean)	\$505.00	\$583.10	-0.0043	Floor — predicts the mean for every customer
Linear Regression	\$223.34	\$319.21	0.6990	
Random Forest	\$162.62	\$253.53	0.8101	Selected — best R ²
Gradient Boosting	\$177.59	\$265.15	0.7923	

Income alone accounts for 75.5% of feature importance in the winning Random Forest model. The model is largely an income lookup — which is honest but means most CLV 'prediction' reduces to: richer customers spend more.

Response classifier — model selection

Target: Response (1 = accepted latest campaign, 0 = did not). Class imbalance: ~15% positive. All three classifiers were trained with consistent class balancing via `sample_weight` (`compute_sample_weight` at fit time) — this ensures the AUC-ROC comparison is apples-to-apples. Gradient Boosting doesn't support `class_weight='balanced'` directly, which is why we used `sample_weight` instead of the `class_weight` parameter.

Model selection by per-class recall, not accuracy. For a campaign-targeting use case, missing a responder is a direct revenue loss. We rank candidates by Negative (Respond) recall first, AUC-ROC as a tiebreaker. Accuracy on a 15/85 imbalance is dominated by the majority class and is not an appropriate selection metric.

Model	Recall (Respond)	Recall (Not)	Accuracy	AUC-ROC	Decision
Baseline (random)	14.9%	84.6%	74.0%	0.4975	Floor
Logistic Regression	80.6%	80.1%	80.1%	0.8833	Selected — highest Respond recall
Random Forest	32.8%	97.9%	88.0%	0.8656	Rejected — highest accuracy, catches only 33% of responders

Gradient Boosting	71.6%	85.4%	83.3%	0.8765	
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Random Forest would have been selected by accuracy alone (88.0%) — but it catches only 1 in 3 actual responders (32.8%). Logistic Regression catches 80.6% of responders at the cost of more false positives. When contacting a few extra non-responders is cheap but missing a real responder costs revenue, this is the correct trade-off.

Targeting priority matrix

CLV tier	Customers	Avg spend	Actual response rate	Predicted probability
High value	737	\$1,355	25.10%	46.90%
Mid value	737	\$416	12.35%	30.34%
Low value	738	\$51	7.72%	21.21%

Calibration caveat. Predicted probabilities (e.g. 46.9% for high-value tier) are higher than actual response rates (25.1%) because the classifier was trained with balanced sample weights — this maximises recall on the minority class but inflates predicted probability values. Use the model for ranking customers, not as a calibrated probability forecast. To get calibrated probabilities, wrap the base model in `CalibratedClassifierCV` with `method='sigmoid'`, but note this will reduce recall on the Respond class.

Phase 5 — Sentiment Analysis

Objective

Build a text classifier that identifies customer complaints (Negative reviews) from Amazon product review text. The business question drives model selection: 'which model best identifies complaints?' means Negative-class recall is the primary ranking metric.

NLP pipeline

Step	Detail	Why
Text preprocessing	Lowercase, strip URLs and punctuation, tokenise, remove stopwords (English + domain-specific), Porter stemmer	Normalise vocabulary, reduce noise
Domain stopwords	Removed: product, item, bought, buy, amazon, price, shipping, arrived, etc.	These appear in every review regardless of sentiment — no discriminating signal
TF-IDF vectorisation	max_features=10,000; ngram_range=(1,2); min_df=3; sublinear_tf=True	Unigrams + bigrams capture phrases like 'dont last'; sublinear dampens very frequent terms
Train/test split	80/20, stratified on sentiment class	Preserve the 90/5/5 class distribution in test set
Class balancing	class_weight='balanced' on Logistic Regression and Linear SVC	Compensate for 90% Positive majority; otherwise models predict Positive for everything

Evaluation order

Models were evaluated in strict order: (1) confusion matrices for all three models to see where errors occur before any summary metric, (2) full classification report per model showing per-class precision, recall, F1, (3) AUC-ROC last as a summary discrimination score. Accuracy was deliberately placed last in the comparison table — on a 90/5/5 imbalance, accuracy is dominated by the Positive majority and is not a useful selection metric.

Model comparison

Model	Neg recall	Neu recall	Pos recall	Accuracy	AUC-ROC	Decision
Naive Bayes	56.7%	12.8%	99.2%	92.9%	0.9282	Misses 43% of complaints
Logistic Regression	85.3%	64.3%	90.7%	89.2%	0.9306	Selected
Linear SVC	69.8%	35.2%	99.2%	94.6%	0.9264	Highest accuracy — misses 30% of complaints

Linear SVC has the highest accuracy (94.6%) but Logistic Regression was selected because it catches 85.3% of complaints vs SVC's 69.8% — a 15.4 percentage-point gap, meaning the selected model finds roughly one in six additional complaints. For a complaint-identification use case, this is the correct trade-off: missing complaints is far more costly than flagging a few neutral reviews as negative.

Qualitative insight — dominant complaint theme

The Negative class TF-IDF top words tell a single clear story: battery, last, long, dead, 'dont last', week, return, disappoint. Amazon product complaints in this dataset are overwhelmingly about battery life, not delivery, packaging, or price. If these products were under the team's control, battery durability would be the single highest-impact product quality fix.

The Neutral class shares vocabulary heavily with Negative (battery, last, dont, work, slow) — Neutral reviews are essentially 'concerns without anger'. This is why Neutral is the hardest class to classify (64.3% recall even

for the selected model), and it's not a model failure — it's a feature of how customers write.

Category breakdown

Category	Reviews	Negative %	Implication
Health & Beauty	9,465	11.52%	4× the Electronics rate — battery-powered beauty devices likely culprit
Toys & Games, Electronics	1,663	3.25%	
Electronics	13,955	2.65%	Large volume, lower relative complaint rate

Standalone scope. These findings reflect reviewer sentiment for Amazon products — not the campaign customers in the main dataset. The two datasets share no join key.

Phase 6 — Market Basket Analysis

Objective

Identify which product categories tend to be purchased by the same customers, and how those patterns differ across CLV tiers from Phase 4.

Critical methodology decision — aggregate spend, not transactions

Classical market basket analysis (Apriori, FP-Growth) is designed for transaction-level data: one row per shopping trip listing the items in it. The spending data is aggregate 2-year spend per customer per category — six dollar amounts per customer, with no individual transactions.

A naive translation ('customer spent > \$0 on wines' = wines in basket) breaks the analysis: over 2 years almost every customer buys at least a little of almost every category, and Apriori would report that essentially every pair co-occurs — technically valid, analytically useless.

What we did instead. We converted continuous spend into a binary 'above-median buyer' signal per category: a customer is treated as a buyer of category X if they spent above the median for that category over the 2-year window. This gives a 50/50 split per category and lets Apriori surface meaningful purchasing affinities — which categories tend to attract the same customer — which is what's actionable for cross-sell.

Algorithm choice — Apriori + FP-Growth cross-check

Apriori was the primary algorithm (min support 10%, min lift 1.2, min confidence 0.5). FP-Growth was run independently on the same data as a cross-check — both algorithms converged on identical frequent itemsets, confirming the results are not algorithm-specific. Per-tier analysis used a relaxed threshold (min support 15%) to account for smaller sub-populations (~740 customers per tier). The above-median signal was recomputed within each tier, not globally, so Low tier results reflect 'above-median for low spenders' rather than a global threshold that would leave them with almost no positives.

Key findings

Finding	Detail
Fish is the connector category	Appears in nearly every top rule across all tiers and the catalog-heavy segment. Customers who buy above-median Fish also tend to buy above-median Fruits, Sweets, and Meat.
Tier-inversion finding	Basket affinity weakens as CLV rises (Mid: 108 rules, max lift 1.87; Low: 55 rules, max lift 1.85; High: 25 rules, max lift 1.42). High spenders buy above-median across all categories — no cross-sell signature.
Catalog-heavy = fresh-food affinity	Top rule among catalog-heavy customers: Fish + Fruits → Meat + Sweets (lift 1.69). Wines and Gold absent. The luxury catalog channel skews toward fresh-food bundles, not premium categories.
High value tier absent from heatmap	None of the 25 High value rules rank in the top 15 overall — their associations are too weak. This is the finding, not a data gap.

Recommendations

- **Cross-sell campaigns — target Mid value customers first.** They show the strongest, most distinctive category affinities (108 rules, max lift 1.87). Lead with Fish-anchored bundles (Fish + Fruits, Fish + Sweets).
- **Catalog-channel campaigns — design a fresh-food bundle.** Fish + Fruits + Meat + Sweets. Specific to catalog-heavy buyers and aligns with the Phase 3 catalog-as-luxury-channel finding.
- **High value customers — avoid product-affinity cross-sell.** They buy everything. Marketing to them should focus on volume incentives, exclusivity, or loyalty mechanics rather than 'you bought X, try Y.'

Dashboard — Tableau Public

Structure

Six dashboards published to Tableau Public, connected to pre-exported CSVs (no live database connection — Tableau Public doesn't support PostgreSQL).

Dashboard	Key charts	Primary finding shown
1A — Campaign Acceptance & Efficiency	KPI strip + acceptance rate bars + break-even analysis	Latest campaign only profitable one; Campaign 2 needs 73.7× revenue to break even
1B — Customer Profile	Responder profile, channel by tier, education/marital response rates	Responders spend 82.6% more; catalog 20× gap; PhD 21% vs Basic 3.7%
2A — CLV Prediction	Model comparison table, CLV tier profile, actual vs predicted scatter, feature importance	Random Forest $R^2=0.81$; income alone 75.5% of importance
2B — Targeting & Propensity	Classifier comparison, predicted vs actual response by tier	Logistic Regression catches 80.6% of responders; High value 3.3× Low value response rate
3 — Sentiment Analysis	Distribution, model comparison, top words by sentiment, negative rate by category	Logistic Regression catches 85.3% of complaints; battery life dominant; H&B; 11.5% negative
4 — Market Basket	Tier summary, strongest pairings, heatmap by tier, catalog-heavy rules	Tier inversion: affinity weakens as CLV rises; Fish is the connector; catalog = fresh-food

Design decisions

- All dashboards use a consistent colour scheme: teal/green/blue tiers, green=positive/selected, red/orange=cautionary, matching across pages.
- Model comparison tables place the most important metric first (Negative Recall for sentiment, Recall Respond for propensity) to guide the reader's eye to the right number before they see accuracy.
- Break-even analysis uses a traffic-light colour bucket (green $\leq 10\times$, yellow $\leq 25\times$, red $> 25\times$) to make the profitability gradient immediately readable without reading axis labels.
- Navigation buttons on every dashboard allow non-linear exploration across all 6 pages.
- Heatmap for market basket uses Orange-Blue Diverging palette centred at lift=1.0 — below-chance associations appear orange, above-chance appear blue.
- Calibration caveat added to the targeting dashboard: predicted probabilities inflate actual response rates due to sample-weight balancing. The model should be used for ranking, not as a literal forecast.

Cross-cutting methodology notes

Model evaluation order (standing rule)

All classification models in this project follow the same evaluation order: (1) confusion matrix — where are the errors? (2) classification report — per-class precision, recall, F1 (3) AUC-ROC — overall discrimination quality, last. This order is intentional: summary metrics like accuracy and AUC can hide class-level failures on imbalanced data. The confusion matrix forces you to see the failures first.

The accuracy trap — caught three times

The same model-selection error appeared in three separate phases and was corrected each time:

Phase	Wrong choice (by accuracy)	Correct choice (by relevant metric)	Gap
Phase 4 — Propensity	Random Forest (88.0% accuracy, 32.8% Respond recall)	Logistic Regression (80.1% accuracy, 80.6% Respond recall)	47.8 pp recall gap
Phase 5 — Sentiment	Linear SVC (94.6% accuracy, 69.8% Neg recall)	Logistic Regression (89.2% accuracy, 85.3% Neg recall)	15.4 pp recall gap

Leakage prevention

Two leakage risks were identified and addressed:

- **CLV regression.** Purchase counts and per-category spend excluded as features — they measure the same behaviour as the target (total_spend) in different units. Using them would inflate R^2 without real predictive ability.
- **Response propensity.** Total_Hist_Campaigns_Accepted sums only AcceptedCmp1–5 — Response (the target) is never included in any feature.

Reproducibility

All modelling code uses random_state=42 for train/test splits and tree-based models. The age reference year is derived empirically from max(Dt_Customer) rather than hardcoded. SQL NTILE(3) is used for tier splits in both the database and the export scripts, ensuring Tableau dashboards show the same tier definitions as the modelling notebooks.

Results summary

Business question	Answer	Key number
Which campaigns generate positive ROI and which customers respond?	Latest campaign is the only one that breaks even under realistic assumptions. Responders are identifiable by past acceptance, spend level, and catalog usage.	15.05% acceptance; 82.6% spend lift; catalog 20× gap
Which customers are most valuable and likely to respond to future campaigns?	High value tier (avg spend \$1,355) responds at 3.3× the rate of low value (\$51). Income is the dominant CLV driver.	R ² =0.81; High tier 25.1% actual response vs 7.7% low
Which product categories are purchased together?	Fish is the connector category across all tiers. Cross-sell is most effective on Mid value customers — High value customers buy everything already.	Mid tier: 108 rules, lift 1.87; High tier: 25 rules, lift 1.42
Which sentiment model best identifies complaints?	Logistic Regression catches 85.3% of complaints. Battery life is the dominant complaint theme. Health & Beauty has the highest complaint rate at 11.52%.	85.3% Neg recall vs 69.8% for Linear SVC